

Transformation Laws and Localized Stability of Generator-Resolved Diagnostics

*Exact Transport Covariance, Carrier-Localized Perturbation, and Minimax Information
Bounds*

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Paper XXV of the RIME program. This paper develops a finite-dimensional methods layer for aligned generator-resolved diagnostics, separating exact transport covariance, carrier-local perturbation, and application-specific semantics.

Abstract

Problem. A generator-resolved block $Q_i Y_a Q_j$ depends on the labelled operator, the marked source and target carriers, and the frame in which those data are compared. Global norm budgets remain portable but can erase exactly the localization information needed to decide whether a thresholded block is stable.

Approach. A typed finite-dimensional diagnostic instance separates representation-intrinsic, labelled-generator-relative, and marked-sectorization-relative data. After two instances are aligned by a unitary, each block difference resolves into left-carrier motion, additive operator error, and right-carrier motion. Global, semi-local, and local error summaries are ordered by an explicit information lattice.

Results. Simultaneous unitary transport preserves weighted averages, commutants, labelled block norms, and thresholded activity relations. The aligned block difference has an exact three-term decomposition with a portable global bound and a sharper carrier-localized bound. All three one-axis global constants are sharp. In the fixed-projector additive-error submodel, global norms cannot detect localization, and every node of the carrier-information lattice has an exact minimax envelope. Additive error has an exact conditional completion for fixed carrier motion, and the localized triangle bound has a complete equality criterion. Margin theorems separate stable activity, stable inactivity, and an information-theoretically complete unresolved interval. The algebra and localized bound extend to bounded oblique carrier resolutions, with explicit projector-conditioning factors. Exact rational controls and bounded float64 observations are kept distinct: Rubik supplies the simultaneous-transport hostile control, while the two-state Markov construction supplies a portability example.

Boundary. The elementary conjugation and telescoping identities are not claimed as new in isolation. The results control declared finite-dimensional blocks; they do not reconstruct a representation, decide survival of an arbitrary routed product, estimate distance to the incidence locus $AB = 0$, or establish general spectral, pseudospectral, Markov-mixing, continuum, or open-quantum-channel stability.

Keywords: operator perturbation; sector projectors; localized matrix blocks; minimax bounds; oblique projections; threshold stability; typed diagnostics.

Notation Table

Symbol	Meaning
V	finite-dimensional complex Hilbert space
A, I	labelled generator and sector index sets
Y_a, Q_i	represented operator and marked sector projector
$\mu_a, M_\mu(Y)$	labelled weight and weighted generator average
$B_a(i, j) = Q_i Y_a Q_j$	generator-resolved block
$\mathcal{R}_\tau(a; i, j)$	thresholded labelled activity predicate
$U, \bar{Y}_a, \bar{Q}_i$	alignment unitary and transported reference data
E_a, Δ_i	operator and carrier perturbations
$D_a(i, j)$	aligned generator-resolved block difference
$b_{\text{loc}}, b_{\text{glob}}, b_{\text{obl}}$	localized, global, and oblique-carrier bounds
$\mathcal{J}_G, \mathcal{J}_L, \mathcal{J}_R, \mathcal{J}_{LR}, \mathcal{J}_{\text{loc}}$	cumulative carrier-information levels
$U_G, U_L, U_R, U_{LR}, U_{\text{loc}}$	exact minimax envelopes at those levels
τ, b	activity threshold and certified error radius

1 Introduction

This paper studies one concrete comparison question: which properties of a labelled operator-block diagnostic survive aligned change of frame, and what is the sharpest block-stability statement justified by a declared level of global or carrier-local information?

The basic coordinate is

$$B_a(i, j) = Q_i Y_a Q_j.$$

It has three owners. The represented family controls Y_a ; the labelled generator or measure policy controls which operators are retained and how they are averaged; and the marked sectorization controls the source and target carriers. A comparison that transports only one owner is not an aligned comparison. A bound that retains only $\|Y_a\|$ and $\|E_a\|$ is valid but may discard whether the perturbation reaches the selected carriers at all.

The theorem spine follows this distinction. Section 2 introduces the typed diagnostic instance, proves simultaneous-transport covariance, and states its fixed-frame and similarity boundaries. Section 3 proves the exact three-term perturbation identity, global and localized bounds, the bounded oblique extension, sharpness, the carrier-information lattice, exact minimax envelopes in the fixed-projector additive-error submodel, conditional additive completion for fixed carrier geometry, and the equality criterion. Section 4 converts certified block radii into a complete threshold trichotomy. Section 5 gives a deliberately narrow two-state Markov portability lift. Section 6 separates exact certificates from bounded observations and states their frozen numerical protocols.

The central distinction is

$$\text{portable global control} \neq \text{carrier-local information} \neq \text{application-specific semantics}.$$

The first inequality creates the minimax information problem. The second prevents a Hilbert-space block estimate from being renamed a probability, support, reconstruction, or routed-composition theorem without additional hypotheses.

Related Work and Novelty Boundary

Norm inequalities, invariant subspaces, oblique projections, and matrix perturbation theory are standard background [5, 10, 6]. Finite Markov-chain hitting times, regeneration, and stationary distributions are standard tools [7, 9, 8]; the narrow lift in Section 5 is stated only under its

explicit two-state hypotheses. The ideal-property estimate for the Frobenius norm, unitary invariance, and the algebraic telescoping of a three-factor difference are standard ingredients; the telescoping inequality alone is not the principal novelty. The contribution is their typed ownership interface: transformation laws for labelled diagnostics, localized evidence resolution, exact minimax envelopes, sharpness and equality theorems, and a complete threshold-uncertainty boundary.

Within the RIME program, Paper II introduced the direct sector block $Q_i Y_a Q_j$ as the carrier of generator transport and related off-diagonal Frobenius mass to sector non-invariance [3]. Paper III separated direct support from projected composition and isolated the static obstruction

$$AB = 0 \iff \text{im } B \subseteq \ker A$$

for a two-block route [4]. Paper VII developed that obstruction into rank-stratified, carrier-resolved incidence geometry; its transported-frame covariance proposition also records covariance of routed products, Frobenius norms, and profile counts under simultaneous transport of operators, sectors, and carriers [2]. Paper XX organizes the fixed-carrier geometry at every finite route depth through carrierwise factorization, cutwise image–kernel tests, and survivor recursion [1].

Theorem 2.1 packages the covariance of Paper II’s direct block at the typed diagnostic level, including labelled weights, weighted averages, commutants, unitarily invariant block norms, and thresholded labelled activity. Its elementary block-conjugation identity is not reclaimed as a new fact. Theorem 3.1 has a different role: it compares one block in two aligned frames and resolves its finite difference into motion of the left carrier, operator, and right carrier. It is a metric finite-difference counterpart of the earlier static block-incidence geometry, not a derivation from the image–kernel criterion.

The perturbation theorem therefore does not decide survival of a routed product, estimate distance to $\{(A, B) : AB = 0\}$, or propagate error through Paper XX’s all-depth survivor recursion. Those extensions require productwise, transversality, or carrier-preservation hypotheses not assumed here.

2 Typed Unitary Transport

2.1 Typed diagnostic instances

Let V be a finite-dimensional complex Hilbert space. Let A and I be finite label sets. A typed generator-resolved diagnostic instance consists of

$$\mathcal{D} = (V, A, I, Y, Q, \mu, \|\cdot\|_F, \text{zero}),$$

where

$$Y : A \longrightarrow \text{End}(V), \quad a \mapsto Y_a, \quad Q : I \longrightarrow \text{End}(V), \quad i \mapsto Q_i,$$

the Q_i are nonzero mutually orthogonal projectors with $\sum_i Q_i = I_V$, and μ is a declared labelled weight family. Labels are part of the data: equal represented operators do not identify distinct source labels.

Define the weighted generator average and labelled blocks by

$$M_\mu(Y) = \sum_{a \in A} \mu_a Y_a, \quad B_a(i, j) = Q_i Y_a Q_j.$$

For a declared threshold $\tau \geq 0$, define the generator-resolved activity predicate

$$\mathcal{R}_\tau(a; i, j) \iff \|B_a(i, j)\|_F > \tau.$$

At $\tau = 0$, this is exact operator nonvanishing. At positive τ , it is a strength policy and must not be renamed exact support.

2.2 Typed Unitary Transport Theorem

Theorem 2.1 (Typed Unitary Transport).

Let

$$\begin{aligned}\mathcal{D} &= (V, A, I, Y, Q, \mu, \|\cdot\|_F, \text{zero}), \\ \mathcal{D}' &= (V', A, I, Y', Q', \mu, \|\cdot\|_F, \text{zero})\end{aligned}$$

have the same generator labels, sector labels, labelled weights, and zero policy. Suppose $U : V \rightarrow V'$ is unitary and, for every $a \in A$ and $i \in I$,

$$Y'_a = UY_aU^*, \quad Q'_i = UQ_iU^*.$$

Then:

1. Q' is again a complete orthogonal sectorization with the same labelled sector dimensions.
2. The weighted averages satisfy

$$M_\mu(Y') = UM_\mu(Y)U^*.$$

Consequently their characteristic polynomials, spectra with algebraic multiplicity, and trace moments agree.

3. Every labelled block satisfies

$$Q'_iY'_aQ'_j = U(Q_iY_aQ_j)U^*.$$

4. For every unitarily invariant matrix norm $\|\cdot\|$,

$$\|Q'_iY'_aQ'_j\| = \|Q_iY_aQ_j\|.$$

In particular, the full Frobenius block profile and every thresholded labelled activity relation $\mathcal{R}_\tau$ are preserved.

5. Conjugation by U is a vector-space isomorphism

$$\text{Comm}(Y) \longrightarrow \text{Comm}(Y'), \quad T \longmapsto UTU^*,$$

so the simultaneous commutant dimensions agree. If the declared family is accompanied by a completeness witness for a represented action, this also preserves the corresponding representation-intrinsic commutant dimension.

Proof. Unitary conjugation preserves adjoints, products, sums, identity, rank, and orthogonality. Therefore

$$Q'_iQ'_j = UQ_iQ_jU^*, \quad \sum_i Q'_i = U \left(\sum_i Q_i \right) U^* = I_{V'}.$$

Linearity gives the weighted-average identity. Associativity and $U^*U = I$ give

$$Q'_i Y'_a Q'_j = U Q_i U^* U Y_a U^* U Q_j U^* = U(Q_i Y_a Q_j) U^*.$$

Unitary invariance gives the norm equality. Hence zero/nonzero and any common threshold comparison are preserved coordinatewise. Finally,

$$T Y_a = Y_a T \iff (U T U^*) Y'_a = Y'_a (U T U^*)$$

for every label a , and the inverse commutant map is conjugation by U^* . This proves all claims.

Corollary 2.2 (Aggregate activity covariance).

Any aggregate relation formed from the labelled predicates by fixed Boolean operations, including

$$(i, j) \in \mathcal{R}_\tau^{\text{agg}} \iff \exists a \in A : \mathcal{R}_\tau(a; i, j),$$

is preserved under the simultaneous transport in Theorem 2.1.

Boundary 2.3 (Fixed-frame hostile control). *If the operators are transported but the projectors are held fixed, the hypotheses of Theorem 2.1 are not satisfied. No block-profile or activity invariance follows. This is an alignment failure, not a counterexample to the theorem.*

Boundary 2.4 (General similarity is algebraic, not metric). *If S is only invertible, the algebraic identities still hold with S^{-1} in place of U^* :*

$$Y'_a = S Y_a S^{-1}, \quad Q'_i = S Q_i S^{-1}, \quad B'_a(i, j) = S B_a(i, j) S^{-1}.$$

But the transported idempotents need not be self-adjoint, and Frobenius block norms need not be preserved. Thus general similarity does not inhabit the metric conclusion of Theorem 2.1 without an additional transported metric.

3 Carrier-Localized Perturbation Theorem

Transport the reference instance into one common Hilbert space and write

$$\bar{Y}_a = U Y_a U^*, \quad \bar{Q}_i = U Q_i U^*.$$

Let the observed operators and projectors be

$$Y'_a = \bar{Y}_a + E_a, \quad Q'_i = \bar{Q}_i + \Delta_i,$$

where both $\bar{Q}$ and Q' are complete orthogonal sectorizations. No normality or unitarity assumption is imposed on Y_a or Y'_a .

Theorem 3.1 (Carrier-Localized Perturbation Bound).

For every $a \in A$ and $i, j \in I$, define

$$D_a(i, j) = Q'_i Y'_a Q'_j - \bar{Q}_i \bar{Y}_a \bar{Q}_j.$$

Then the exact three-term identity

$$D_a(i, j) = \Delta_i \bar{Y}_a Q'_j + Q'_i E_a Q'_j + \bar{Q}_i \bar{Y}_a \Delta_j$$

holds. Consequently,

$$\|D_a(i, j)\|_F \leq b_{\text{loc}}(a; i, j),$$

where

$$b_{\text{loc}}(a; i, j) = \|\Delta_i \bar{Y}_a Q'_j\|_F + \|Q'_i E_a Q'_j\|_F \\ + \|\bar{Q}_i \bar{Y}_a \Delta_j\|_F.$$

If declared bounds satisfy

$$\|\Delta_k\|_F \leq \delta_k, \quad \|E_a\|_F \leq \varepsilon_{a,F},$$

then also

$$\|D_a(i, j)\|_F \leq b_{\text{glob}}(a; i, j),$$

with

$$b_{\text{glob}}(a; i, j) = \|Y_a\|_2(\delta_i + \delta_j) + \varepsilon_{a,F}.$$

Moreover, when the right-hand quantities are evaluated from the same exact operators and declared upper bounds,

$$b_{\text{loc}}(a; i, j) \leq b_{\text{glob}}(a; i, j).$$

Proof. First separate the additive error:

$$Q'_i Y'_a Q'_j - \bar{Q}_i \bar{Y}_a \bar{Q}_j = [Q'_i \bar{Y}_a Q'_j - \bar{Q}_i \bar{Y}_a \bar{Q}_j] + Q'_i E_a Q'_j.$$

Insert and subtract $\bar{Q}_i \bar{Y}_a Q'_j$ in the bracket. The three differences are exactly the displayed terms. The triangle inequality gives b_{loc} .

For the global estimate, orthogonal projectors have operator norm at most one, unitary transport preserves $\|Y_a\|_2$, and compression by two orthogonal projectors cannot increase Frobenius norm. Using $\|AB\|_F \leq \|A\|_F \|B\|_2$ and $\|AB\|_F \leq \|A\|_2 \|B\|_F$ bounds the three localized terms respectively by

$$\delta_i \|Y_a\|_2, \quad \varepsilon_{a,F}, \quad \|Y_a\|_2 \delta_j.$$

Their sum is b_{glob} , and the same termwise estimates prove $b_{\text{loc}} \leq b_{\text{glob}}$.

Extension 3.1a (Bounded oblique carriers). The exact three-term identity and the inequality

$$\|D_a(i, j)\|_F \leq b_{\text{loc}}(a; i, j)$$

do not require self-adjointness, mutual orthogonality, or idempotence. They only use

$$Q'_k = \bar{Q}_k + \Delta_k, \quad Y'_a = \bar{Y}_a + E_a.$$

To retain carrier semantics, suppose now that $\bar{Q}_k$ and Q'_k are bounded, possibly oblique idempotents belonging to two declared resolutions of the identity. Put

$$\bar{q}_k = \|\bar{Q}_k\|_2, \quad q'_k = \|Q'_k\|_2, \quad \eta_k = \|\Delta_k\|_2, \quad F_a = \|\bar{Y}_a\|_F, \quad \varepsilon_{a,F} = \|E_a\|_F.$$

Then the same identity gives the portable oblique estimate

$$\|D_a(i, j)\|_F \leq b_{\text{obl}}(a; i, j) = \eta_i F_a q'_j + q'_i \varepsilon_{a,F} q'_j + \bar{q}_i F_a \eta_j.$$

In particular, if all four relevant carrier idempotents have operator norm at most K , then

$$b_{\text{obl}}(a; i, j) \leq KF_a(\eta_i + \eta_j) + K^2 \varepsilon_{a,F}.$$

If the reference carriers are orthogonal, then $\bar{q}_i = \bar{q}_j = 1$ and $q'_k \leq 1 + \eta_k$, which gives the fully delta-controlled specialization

$$b_{\text{obl}}(a; i, j) \leq F_a[\eta_i(1 + \eta_j) + \eta_j] + (1 + \eta_i)(1 + \eta_j)\varepsilon_{a,F}.$$

Proof. The algebraic decomposition and triangle inequality are unchanged. Apply the ideal-property estimate

$$\|ABC\|_F \leq \|A\|_2 \|B\|_F \|C\|_2$$

to the three summands. This gives, in order,

$$\eta_i F_a q'_j, \quad q'_i \varepsilon_{a,F} q'_j, \quad \bar{q}_i F_a \eta_j.$$

The two specializations follow by substitution and by $\|\bar{Q}_k + \Delta_k\|_2 \leq 1 + \eta_k$ in the orthogonal-reference case.

This extension is a stability statement, not a similarity invariance statement: unlike Theorem 2.1, it does not assert preservation of Frobenius block norms. The factors $\bar{q}_k, q'_k$ explicitly record the conditioning cost of passing from orthogonal sectors to oblique carrier coordinates.

Remark 3.2 (Why the additive error must be combined first).

The alternative split

$$\Delta_i Y'_a Q'_j + \bar{Q}_i E_a Q'_j$$

is algebraically valid, but separately bounding its two terms double-counts the part $\Delta_i E_a Q'_j$. It yields the weaker global expression

$$\delta_i(\|Y_a\|_2 + \varepsilon_{a,2}) + \varepsilon_{a,F} + \|Y_a\|_2 \delta_j.$$

Theorem 3.1 combines those terms as $Q'_i E_a Q'_j$ and removes the spurious $\delta_i \varepsilon_{a,2}$ contribution. The weaker expression is therefore not minimax optimal under the stated global information.

Theorem 3.3 (One-Axis Sharpness of the Global Constants).

Each of the three coefficients in

$$b_{\text{glob}} = \|Y_a\|_2 \delta_i + \varepsilon_{a,F} + \|Y_a\|_2 \delta_j$$

is best possible when the other two perturbation axes are zero. More precisely:

1. for every $\varepsilon \geq 0$, there is a fixed-sectorization additive perturbation with $\|E_a\|_F = \varepsilon$ and block difference exactly ε ;
2. for every $M \geq 0$ and $\delta \geq 0$, there is a finite-dimensional left-projector perturbation with $\|Y_a\|_2 = M$, $\|\Delta_i\|_F = \delta$, and block difference exactly $M\delta$;
3. the analogous statement holds for a right-projector perturbation.

Consequently no one of the constants 1, 1, 1 can be uniformly decreased in a global-budget theorem, even though no claim is made that the sum is the exact joint minimax value when several axes are simultaneously nonzero.

Proof. For the additive axis, take fixed coordinate projectors onto $\mathbb{C}e_1$ and $\mathbb{C}e_2$, let $\bar{Y}_a = 0$, and set $E_a = \varepsilon|e_1\rangle\langle e_2|$.

For the left-projector axis, choose an integer r with $\delta \leq \sqrt{2r}$. Let A, B, C be mutually orthogonal subspaces with dimensions $r, r, 2r$. Let $\bar{Q}_i$ project onto A , let an auxiliary sector project onto B , and let $\bar{Q}_j = Q'_j$ project onto C . Rotate each paired coordinate plane of $A \oplus B$ through an angle θ satisfying

$$\sin \theta = \frac{\delta}{\sqrt{2r}}.$$

The resulting rank- r projector Q'_i satisfies $\|Q'_i - \bar{Q}_i\|_F = \sqrt{2r} \sin \theta = \delta$. The self-adjoint projector difference has $2r$ singular values equal to $\sin \theta$. Choose $\bar{Y}_a$ to be M times an isometry from C onto a complete right-singular basis of that difference and zero on $C^\perp$. Then

$$\|(Q'_i - \bar{Q}_i)\bar{Y}_a Q'_j\|_F = M\delta.$$

Completing Q'_i by the rotated auxiliary projector gives a valid complete orthogonal sectorization. Taking adjoints gives the right-projector example. This proves all three assertions.

Theorem 3.4 (Global Information Cannot Detect Carrier Localization).

Fix coordinate projectors $Q_i = |e_1\rangle\langle e_1|$ and $Q_j = |e_2\rangle\langle e_2|$ in a carrier of dimension at least three, and set $\bar{Y}_a = 0$. For $\alpha \in [0, 1]$, define

$$u_\alpha = \alpha e_1 + \sqrt{1 - \alpha^2} e_3, \quad E_\alpha = \varepsilon |u_\alpha\rangle\langle e_2|.$$

All members of this family have exactly the same global data

$$\|E_\alpha\|_2 = \|E_\alpha\|_F = \varepsilon, \quad \|Y_a\|_2 = \delta_i = \delta_j = 0,$$

but

$$\|Q_i E_\alpha Q_j\|_F = \alpha \varepsilon, \quad \frac{b_{\text{loc}}}{b_{\text{glob}}} = \alpha \quad (\varepsilon > 0).$$

Thus $b_{\text{loc}}/b_{\text{glob}}$ can be arbitrarily small and can equal zero. In the fixed-projector additive-error submodel, the single localized seminorm

$$\lambda_{ij}(E_a) = \|Q_i E_a Q_j\|_F$$

is an exact certificate for the block error. Conversely, any information summary that certifies a bound smaller than ε for E_0 must distinguish E_0 from E_1 ; the two global norms alone cannot do so.

Proof. E_α is rank one with its two displayed global norms equal to ε . Its marked compression is $\alpha \varepsilon |e_1\rangle\langle e_2|$. Since the projectors are fixed and the reference operator vanishes, this compression is the complete block difference and is also the sole nonzero term in b_{loc} . At $\alpha = 0$ and $\alpha = 1$ the global records are identical while the exact local errors are respectively zero and ε .

Remark 3.5 (Information ownership).

b_{glob} needs only global norms. b_{loc} needs the three carrier-restricted actions in Theorem 3.1, or just their three Frobenius norms if the actions have already been evaluated. Theorem 3.4 proves a necessity statement in the operator-only submodel: some statistic that separates on-carrier from off-carrier perturbations is required for a strict localized improvement. It does not claim that one unique data structure is minimal among all equivalent encodings.

Definition 3.6 (Cumulative Carrier-Information Lattice).

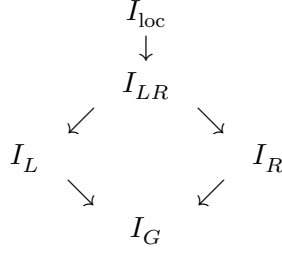
In the fixed-projector additive-error submodel, put $P = Q_i$, $Q = Q_j$, $p = \text{rank } P$, $q = \text{rank } Q$, and $k = \min(p, q)$. Consider certified upper budgets

$$\|E\|_2 \leq \varepsilon_2, \quad \|E\|_F \leq \varepsilon_F, \quad \|PE\|_F \leq \ell, \quad \|EQ\|_F \leq r, \quad \lambda = \|PEQ\|_F.$$

The cumulative information levels are

$$\begin{aligned} I_G &= (\varepsilon_2, \varepsilon_F), \\ I_L &= (I_G, \ell), & I_R &= (I_G, r), \\ I_{LR} &= (I_G, \ell, r), \\ I_{\text{loc}} &= (I_G, \ell, r, \lambda). \end{aligned}$$

They form the refinement lattice



The two semi-local nodes are incomparable: one resolves the target carrier of the perturbation and the other resolves its source carrier.

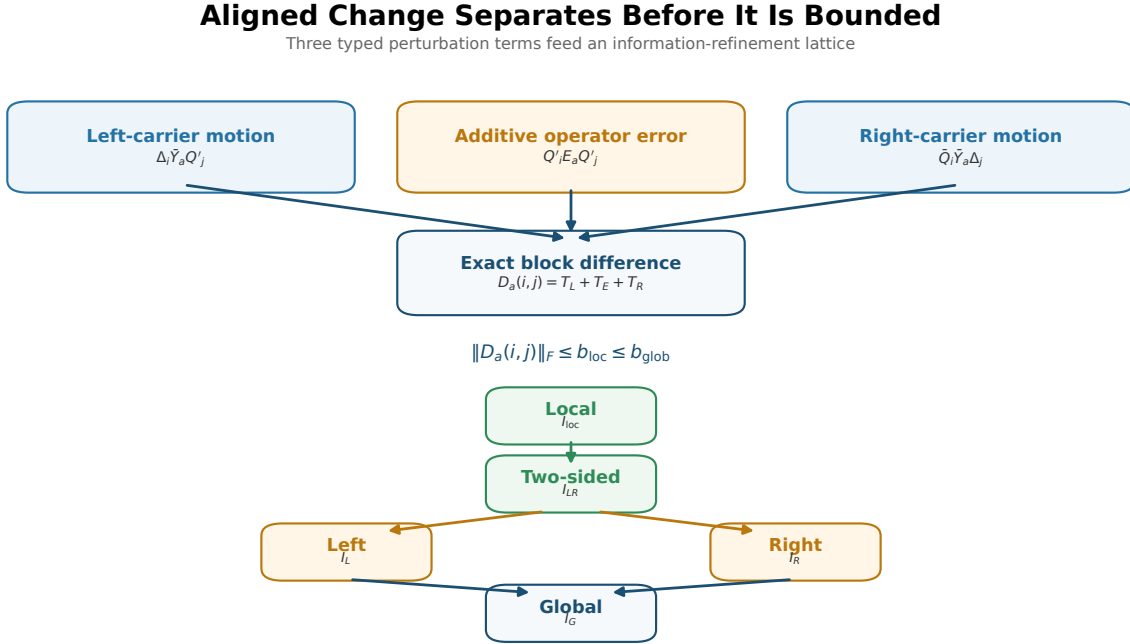


Figure 1. The aligned block difference separates into left-carrier motion, additive operator error, and right-carrier motion. Global and semi-local summaries form an information-refinement lattice whose local node retains the marked block quantity.

Theorem 3.7 (Exact Minimax Envelope at Every Information Level).

Among all operators satisfying the budgets visible at the declared node, the best possible universal upper bounds for $\lambda = \|PEQ\|_F$ are

$$\begin{aligned}
U_G &= \min(\varepsilon_F, \sqrt{k}\varepsilon_2), \\
U_L &= \min(U_G, \ell), \quad U_R = \min(U_G, r), \\
U_{LR} &= \min(U_G, \ell, r), \\
U_{\text{loc}} &= \lambda.
\end{aligned}$$

Every displayed envelope is minimax sharp. When $\dim V \geq 2$ and the marked carriers and their complements are nontrivial, every edge is strict as an information refinement: the statistic added at the upper node is not determined by the lower node. The nodes I_L and I_R are incomparable, and even I_{LR} does not determine λ . This information-theoretic strictness does not assert a strict pointwise inequality between the two envelope values for every numerical budget tuple.

Proof. Compression gives

$$\lambda \leq \|E\|_F, \quad \lambda \leq \|PE\|_F, \quad \lambda \leq \|EQ\|_F.$$

The block PEQ has rank at most k , so

$$\lambda \leq \sqrt{k} \|PEQ\|_2 \leq \sqrt{k} \|E\|_2.$$

Taking the minimum of the inequalities available at each node proves the upper bounds. For sharpness, let m be the displayed minimum and choose orthonormal vectors $p_1, \dots, p_k$ in $\text{ran } P$ and $q_1, \dots, q_k$ in $\text{ran } Q$. Then

$$E = \frac{m}{\sqrt{k}} \sum_{t=1}^k |p_t\rangle\langle q_t|$$

has $\|E\|_F = \lambda = m$, $\|E\|_2 = m/\sqrt{k}$, and both semi-local norms equal to m . It therefore satisfies every visible upper budget and attains the corresponding envelope.

For strictness, choose unit vectors $p_1 \in \text{ran } P$, $p_0 \in \ker P$, $q_1 \in \text{ran } Q$, and $q_0 \in \ker Q$. The rank-one operators

$$E_{11} = |p_1\rangle\langle q_1|, \quad E_{10} = |p_1\rangle\langle q_0|, \quad E_{01} = |p_0\rangle\langle q_1|$$

all have global operator and Frobenius norms one, while their left and right semi-local norms distinguish the two directions independently. Finally, put

$$E_{\text{diag}} = E_{11} + |p_0\rangle\langle q_0|, \quad E_{\text{cross}} = E_{10} + E_{01}.$$

These two operators have the same exact statistics

$$\|E\|_2 = 1, \quad \|E\|_F = \sqrt{2}, \quad \|PE\|_F = \|EQ\|_F = 1,$$

but their local block norms are respectively 1 and 0. Hence every claimed strictness and incomparability already occurs inside the declared nontrivial carrier and complement directions. In particular, the two semi-local statistics together still do not determine λ ; the local node is a genuine strict refinement.

Theorem 3.8 (Exact Additive Completion of Fixed Carrier Motion).

Fix orthogonal projectors $\bar{Q}_i, Q'_i, \bar{Q}_j, Q'_j$ and a reference operator $\bar{Y}_a$. Put

$$A = Q'_i \bar{Y}_a Q'_j - \bar{Q}_i \bar{Y}_a \bar{Q}_j$$

and let

$$\mathcal{S}'_{ij} = \{Q'_i X Q'_j : X \in \text{End}(V)\}.$$

Write the Hilbert–Schmidt orthogonal decomposition

$$A_{\parallel} = Q'_i A Q'_j, \quad A_{\perp} = A - A_{\parallel}.$$

Then, for every $\varepsilon \geq 0$, the exact worst-case block error over additive perturbations with $\|E_a\|_F \leq \varepsilon$ is

$$\sup_{\|E_a\|_F \leq \varepsilon} \|Q'_i(\bar{Y}_a + E_a)Q'_j - \bar{Q}_i \bar{Y}_a \bar{Q}_j\|_F = \sqrt{\|A_{\perp}\|_F^2 + (\|A_{\parallel}\|_F + \varepsilon)^2}.$$

Thus the additive budget combines linearly only with the component of the carrier-motion error already lying in the final block space. It combines orthogonally with the remaining component.

Proof. The map

$$\Pi'_{ij}(X) = Q'_i X Q'_j$$

is the Hilbert–Schmidt orthogonal projector onto S'_{ij} . Moreover, the set of compressed additive errors with $\|E_a\|_F \leq \varepsilon$ is exactly the closed Frobenius ball of radius ε in S'_{ij} : compression cannot increase the norm, while every $C \in S'_{ij}$ is realized by taking $E_a = C$. Therefore, for $C = Q'_i E_a Q'_j$, orthogonality gives

$$\|A + C\|_F^2 = \|A_{\perp}\|_F^2 + \|A_{\parallel} + C\|_F^2.$$

The second term is maximized by choosing C parallel to $A_{\parallel}$; if that component vanishes, any $C \in S'_{ij}$ of norm ε attains the same value. This proves the formula.

Corollary 3.9 (Explicit Three-Axis Strictness).

The three-term localized sum and the scalar global sum are not, in general, the exact joint optimum for a fixed carrier geometry. Let e_1, e_2, e_3, e_4 be orthonormal, choose $c, s \in (0, 1)$ with $c^2 + s^2 = 1$, and set

$$\begin{aligned} \bar{Q}_i &= |e_1\rangle\langle e_1|, & Q'_i &= |u\rangle\langle u|, & u &= ce_1 + se_3, \\ \bar{Q}_j &= |e_2\rangle\langle e_2|, & Q'_j &= |v\rangle\langle v|, & v &= ce_2 + se_4, \\ \bar{Y}_a &= |e_1\rangle\langle e_2|. \end{aligned}$$

Complete both pairs to orthogonal sectorizations using their orthogonal rotated complements. Then

$$\|\Delta_i\|_F = \|\Delta_j\|_F = \sqrt{2}s, \quad \|\bar{Y}_a\|_2 = 1.$$

For the carrier-motion error A of Theorem 3.8,

$$A_{\parallel} = 0, \quad \|A\|_F^2 = 1 - c^4 = s^2(1 + c^2).$$

Hence, with any $\varepsilon > 0$, all three perturbation axes are nonzero and the true additive worst case is

$$\sqrt{s^2(1 + c^2) + \varepsilon^2}.$$

By contrast, the three-term localized and global bounds are respectively

$$b_{\text{loc}} = cs + s + \varepsilon, \quad b_{\text{glob}} = 2\sqrt{2}s + \varepsilon.$$

Both inequalities are strict. In particular, the sum in Theorem 3.1 is a portable certificate, not a claim of joint minimax optimality.

Proof. Direct multiplication gives $Q'_i \bar{Y}_a Q'_j = c^2 |u\rangle\langle v|$. Its projection against the final block space cancels the projection of $|e_1\rangle\langle e_2|$, so $A_{\parallel} = 0$. The two rank-one operators have Hilbert–Schmidt inner product c^2 , giving

$$\|A\|_F^2 = c^4 + 1 - 2c^4 = 1 - c^4.$$

Theorem 3.8 gives the exact supremum. The first and third localized terms have norms cs and s , and the additive term can have norm ε . Finally, $\sqrt{x^2 + y^2} < x + y$ for positive x, y , while $\sqrt{1 + c^2} < 1 + c < 2\sqrt{2}$. This proves both strict comparisons.

Theorem 3.10 (Equality Criterion for the Localized Bound).

Write the three terms in Theorem 3.1 as

$$T_L = \Delta_i \bar{Y}_a Q'_j, \quad T_E = Q'_i E_a Q'_j, \quad T_R = \bar{Q}_i \bar{Y}_a \Delta_j.$$

Then

$$\|D_a(i, j)\|_F = b_{\text{loc}}(a; i, j)$$

if and only if there are a matrix H with $\|H\|_F = 1$ and nonnegative real numbers $\alpha_L, \alpha_E, \alpha_R$ such that

$$T_L = \alpha_L H, \quad T_E = \alpha_E H, \quad T_R = \alpha_R H.$$

Zero terms correspond to zero coefficients. Equivalently, every two nonzero terms have Hilbert–Schmidt inner product equal to the product of their norms, with positive real phase.

For differences of orthogonal projectors, this condition forces

$$\alpha_L \alpha_E \alpha_R = 0.$$

Thus all three localized terms can never be simultaneously nonzero at an equality point. Equality is possible with one active term, or with two active terms that lie on the same nonnegative ray.

Proof. The Frobenius norm is the Hilbert-space norm associated with the Hilbert–Schmidt inner product. Equality in its finite triangle inequality holds exactly when all nonzero summands lie on one nonnegative ray. This proves the first equivalence.

It remains to use projector geometry. Suppose all three coefficients were positive. Because $T_E = Q'_i T_E Q'_j$, the range of H lies in $\text{ran } Q'_i$. Because $T_R = \bar{Q}_i T_R$, it also lies in $\text{ran } \bar{Q}_i$. Hence

$$\text{ran } H \subseteq \text{ran } Q'_i \cap \text{ran } \bar{Q}_i.$$

The self-adjoint difference $\Delta_i = Q'_i - \bar{Q}_i$ annihilates this intersection. Consequently its range is orthogonal to the intersection:

$$\langle h, \Delta_i x \rangle = \langle \Delta_i h, x \rangle = 0$$

for every h in the intersection. But the range of T_L lies in the range of Δ_i , while $T_L = \alpha_L H$ would put the same nonzero range inside the intersection, a contradiction. Therefore at least one coefficient is zero. The converse follows immediately from the common-ray condition.

Remark 3.11 (When the global bound is also attained).

Equality $\|D_a(i, j)\|_F = b_{\text{loc}} = b_{\text{glob}}$ additionally requires equality in every active submultiplicative estimate used in Theorem 3.1, and requires every declared upper budget appearing in b_{glob} to be saturated:

$$\begin{aligned}\|\Delta_i \bar{Y}_a Q'_j\|_F &= \|\Delta_i\|_F \|Y_a\|_2, \\ \|Q'_i E_a Q'_j\|_F &= \|E_a\|_F, \\ \|\bar{Q}_i \bar{Y}_a \Delta_j\|_F &= \|Y_a\|_2 \|\Delta_j\|_F,\end{aligned}$$

together with the common-ray criterion above. Theorem 3.10 shows that the three displayed equalities cannot all contribute positively to one joint equality point.

4 Margin consequences

Corollary 4.1 (Typed activity margin).

Let

$$n = \|\bar{Q}_i \bar{Y}_a \bar{Q}_j\|_F, \quad n' = \|Q'_i Y'_a Q'_j\|_F,$$

and let b be either valid bound from Theorem 3.1. Then

$$|n' - n| \leq b.$$

For a declared threshold $\tau \geq 0$:

$$\begin{aligned}n - b > \tau &\implies n' > \tau \quad (\text{STABLE_ACTIVE}), \\ n + b \leq \tau &\implies n' \leq \tau \quad (\text{STABLE_INACTIVE}), \\ \text{otherwise} &\quad (\text{UNRESOLVED}).\end{aligned}$$

UNRESOLVED is a proof-status outcome. It is not evidence that the perturbed block is zero, nonzero, active, or inactive.

Theorem 4.2 (Completeness of the Unresolved Interval).

For $n, b, \tau \geq 0$, let the only available scalar error information be

$$n' \geq 0, \quad |n' - n| \leq b.$$

Then the margin policy of Corollary 4.1 returns UNRESOLVED if and only if the admissible interval contains both an inactive value and an active value:

$$\begin{aligned}\exists x_- : \quad &|x_- - n| \leq b, \quad x_- \leq \tau, \\ \exists x_+ : \quad &|x_+ - n| \leq b, \quad x_+ > \tau.\end{aligned}$$

Both alternatives are realized by finite-dimensional operator-only perturbations with fixed coordinate projectors. Hence no sound classifier using only n , b , and τ can refine UNRESOLVED to either stable status.

Proof. *The policy is unresolved exactly when*

$$n - b \leq \tau < n + b.$$

In that case $x_- = \max(0, n - b)$ is admissible and inactive, while any x_+ strictly between τ and $n + b$, for example their midpoint, is admissible and active. Conversely, the existence of an admissible inactive value rules out $n - b > \tau$, and the existence of an admissible active value rules out $n + b \leq \tau$.

For realization, use fixed projectors onto $\mathbb{C}e_1$ and $\mathbb{C}e_2$, take $\bar{Y}_a = n|e_1\rangle\langle e_2|$, and set $Y'_a = x|e_1\rangle\langle e_2|$. Then $E_a = (x - n)|e_1\rangle\langle e_2|$ has Frobenius norm at most b and the perturbed block norm is exactly x . Applying this once with $x = x_-$ and once with $x = x_+$ gives the two hostile realizations.

Corollary 4.3 (Finite aggregate margin).

For a finite coordinate family K , suppose $|n'_k - n_k| \leq b_k$ for every $k \in K$. Then

$$\max_{k \in K} \max(0, n_k - b_k) \leq \max_{k \in K} n'_k \leq \max_{k \in K} (n_k + b_k).$$

The same threshold trichotomy therefore applies to an aggregate coordinate using these lower and upper bounds. Section 6.2 applies this aggregate policy to the Rubik observation; the Markov example in Section 6.3 remains a coordinatewise semantic lift.

5 Two-state Markov probability lift

The following is an application with additional structure, not a conclusion about arbitrary Markov systems.

Corollary 5.1 (Singleton Markov Lift).

Let

$$P = \begin{pmatrix} 1 - a & a \\ \kappa & 1 - \kappa \end{pmatrix}$$

be row-stochastic with $a, \kappa > 0$, and let Q_0, Q_1 be the coordinate singleton projectors. Under the convention $P[\text{source}, \text{target}]$,

$$\|Q_0 P Q_1\|_F = a, \quad \|Q_1 P Q_0\|_F = \kappa.$$

Suppose a perturbation preserves this two-state form and the reverse probability κ , while Theorem 3.1 gives

$$a' \in [\ell, u], \quad \ell = \max(0, a - \beta), \quad u = \min(1, a + \beta).$$

If $\ell > 0$, then throughout the admitted interval:

the positive edge $0 \rightarrow 1$ persists,

$$\Pr_0(T_1 < \infty) = 1,$$

$$\mathbb{E}_0[T_1] \in [1/u, 1/\ell],$$

$$\pi_1 \in \left[\frac{\ell}{\ell + \kappa}, \frac{u}{u + \kappa} \right],$$

$$\text{sep}_1 = |1 - \lambda_2| \in [\kappa + \ell, \kappa + u].$$

Proof. The coordinate block is $a e_0 e_1^*$, whose Frobenius norm is a because $a \geq 0$. For $a' > 0$, the first transition from state 0 to state 1 has a geometric waiting time with parameter a' , hence eventual hit probability one and expectation $1/a'$. Solving $\pi P = \pi$ gives $\pi_1 = a'/(a' + \kappa)$. The eigenvalues are 1 and $\lambda_2 = 1 - a' - \kappa$, so the separation from the stationary eigenvalue is $|1 - \lambda_2| = a' + \kappa$. All displayed functions are monotone on the admitted interval, yielding the endpoint bounds.

If $\ell = 0$, the block theorem alone does not certify edge persistence, almost-sure hitting, or a finite upper hitting-time bound. A localized bound may make ℓ positive when the global bound does not, as in the two-state control registered in Section 6.3.

Boundary 5.2 (Partition and threshold discipline). Corollary 5.1 is not available after continuously rotating the coordinate projectors: the result is still an orthogonal Hilbert frame but no longer a set-valued partition of the Markov states. Extension 3.1a remains algebraically and metrically valid for a bounded oblique carrier resolution, but such a resolution is even further

from a state partition and therefore does not restore the Markov probability interpretation. Also, for any positive threshold τ ,

$$0 < a \leq \tau$$

is a positive Markov edge that remains inactive under the block-strength policy. Positive support and thresholded block activity are different types.

5.1 Supplementary Technical Note S1

The separately packaged note proves one nontrivial proportional-row preservation contract for finite chains: expected hitting time decreases and stationary target mass increases as the direct target probability grows, while the absolute spectral gap need not be monotone. The classification of general preservation contracts remains open.

6 Exact Controls and Bounded Realizations

The manuscript proofs are theorem authority. The computational records are registered in two distinct layers. The exact layer uses integer and `Fraction` arithmetic and literal exact-zero semantics. The bounded layer uses the frozen Python/NumPy float64/complex128 protocols stated below; its records are observations for declared fixtures, not proofs of the general theorems.

6.1 Exact rational controls

The exact certificate records one-axis equality witnesses for all three global constants, the fixed-global-data localization family of Theorem 3.4, the carrier-information lattice and its strict hostile pairs, an exact three-axis witness for Corollary 3.9, a localized-bound equality witness, and two-sided realizations of the unresolved interval. All matrix entries are rational, and the registry records the integer/Fraction backend, norm protocol, threshold policy, and exact-zero policy. It is a finite certificate for the declared fixtures, not a second proof of the general theorems.

The frozen exact protocol uses Python 3.13.0, Python integers and `fractions.Fraction`, exact Frobenius norms through squared entries, and exact witness identities for the operator-norm fields. Its threshold fixture uses $n = b = \tau = 1$, with `UNRESOLVED` defined by $n - b \leq \tau < n + b$. Zero means literal rational equality to 0; no numerical tolerance or near-zero substitution is admitted.

6.2 Rubik simultaneous-transport hostile control

The Rubik control uses the canonical 228-dimensional cubie carrier, all 18 labelled generators, and 9×9 ordered sector coordinates. It tests simultaneous transport of operators and labelled carriers against the hostile fixed-frame control. It does not serve as a general perturbation distribution or Markov example. The frozen bounded protocol uses Python 3.13.0, NumPy 2.1.3, SciPy 1.18.0, complex128/float64 arithmetic, Frobenius block norms, spectral operator norms, Frobenius projector deltas, support thresholds 10^{-10} , 10^{-6} , and 0.05, activity threshold 0.05, transport residual tolerance 10^{-10} , and perturbation comparison tolerance 2×10^{-10} . A localized/global ratio at most the latter tolerance is reported as zero; a threshold-inactive coordinate is not thereby an exact-zero certificate. The generic near-zero control uses a separately declared error radius 10^{-9} .

At the largest declared perturbation level, $\varepsilon = \eta = 0.05$, the following table reports the coordinatewise ratio $b_{\text{loc}}/b_{\text{glob}}$ only where $b_{\text{glob}} > 0$. The two bound maxima are separate maxima over all coordinates; their quotient is not used as a coordinatewise statistic.

axis	positive- global coordinates	zero ratio	median ratio	mean ratio	maximum ratio
operator	1458	1440	0	0.01235	1.00000
sector	576	408	$< 10^{-15}$	0.11972	0.70711
coupled	1458	1290	0	0.03553	0.77006

The corresponding bound maxima and aggregate margin transitions are:

axis	$\max b_{\text{glob}}$	$\max b_{\text{loc}}$	aggregate unresolved, global $\rightarrow$ local
operator	0.05000	0.05000	24 $\rightarrow$ 1
sector	0.14136	0.09996	13 $\rightarrow$ 1
coupled	0.19490	0.15008	24 $\rightarrow$ 1

Thus the localized theorem is not merely a modest improvement of the largest global scalar. In the operator-only control, only the 18 coordinates hit by the rank-one generator errors retain ratio one; 1440 other coordinates have the same positive global budget but localized bound reported as zero under the frozen comparison tolerance. On the coupled axis every coordinate is strictly improved, and the aggregate unresolved census drops from 24 to 1. These are bounded float64 observations for the declared carrier and perturbation family, not universal distribution laws.

6.3 Markov portability and cross-layer example

The Markov record is deliberately narrower. It uses the two-state singleton partition to illustrate how a certified block interval lifts to positive-edge, hitting-time, stationary-mass, and eigenvalue-separation statements under the additional hypotheses of Corollary 5.1. Rotated Hilbert frames appear only as a cross-layer boundary: they are not state partitions, so no Markov probability conclusion is admitted. The frozen bounded protocol uses Python 3.13.0, NumPy 2.1.3, float64/complex128 arithmetic, block-activity threshold 0.15, positive-support threshold 10^{-12} , and comparison tolerance 2×10^{-12} . Positive support and thresholded block activity remain distinct policies; UNRESOLVED is not an exact-zero status.

7 Claim Status and Boundary

The theorem and evidence levels are:

Surface	Claim level	Registered control
Theorem 2.1, Corollary 2.2, and Boundary 2.3	Theorem/boundary	exact integer/Fraction transport fixture and Rubik simultaneous-transport hostile control
Theorem 3.1 and Extension 3.1a	Theorem	Rubik orthogonal-specialization observation; Extension 3.1a is proof-only
Theorems 3.3–3.4	Theorem	exact rational sharpness and separation controls
Definition 3.6 and Theorem 3.7	Theorem	exact rational lattice grid and hostile pairs
Theorem 3.8 and Corollary 3.9	Theorem	exact rational conditional-minimax and three-axis controls
Theorem 3.10	Theorem	exact rational equality witness
Corollary 4.1, Theorem 4.2, and Corollary 4.3	Theorem	exact two-sided hostile realizations and bounded Rubik aggregate sweep
Corollary 5.1	Theorem under its stated two-state hypotheses	bounded Markov portability example

Supplementary Technical Note S1 has its own theorem/boundary ledger and local validator under `experiments/paper25/notes/`. It is part of the Paper XXV release package but not part of the main theorem numbering above.

No claim is made that aggregate block profiles reconstruct the labelled operator family or its representation; that global norms determine carrier localization; that thresholded activity equals exact support; that an orthogonal Hilbert frame or oblique carrier resolution is a set-valued state partition; or that a one-block perturbation estimate controls arbitrary-depth routed composition. Theorem 3.8 is conditional on fixed carrier geometry and does not determine the joint minimax envelope from $\delta_i, \delta_j, \varepsilon_{a,F}, \|Y_a\|_2$ alone across all sectorizations. The Rubik table is a bounded float64 observation for one declared carrier and perturbation family, not a universal distribution law; the Markov observation is confined to the stated two-state model.

8 Conclusion

Aligned unitary transport preserves the complete typed block profile because the represented operators and marked carriers move together. After alignment, the change of one generator-resolved block separates exactly into left-carrier motion, additive operator error, and right-carrier motion. The resulting localized bound retains information that every portable global norm budget forgets.

These results make that information loss quantitative. The global coefficients are sharp one axis at a time. In the fixed-projector additive-error submodel, global norms cannot detect localization and the global/semi-local/local information lattice has an exact minimax envelope at every node. For fixed carrier motion, additive error admits an exact conditional completion, while the localized triangle bound has a complete equality criterion. The threshold interval left unresolved by a certified radius is complete: the admitted information genuinely permits realizations on both sides of the threshold.

The algebra extends beyond orthogonal sectors to bounded oblique carrier resolutions, with explicit conditioning factors. Application semantics do not extend automatically. In particular, the two-state Markov lift requires a genuine singleton state partition, and the one-block theorem does not replace the image–kernel or survivor tests needed for routed composition.

Supplementary Technical Note S1 supplies one positive n-state preservation contract for hitting time and stationary mass together with a negative absolute-gap boundary. It does not close the broader classification problem.

9 Open Problems

The established boundary leaves the following questions open:

1. **Joint oblique minimax geometry.** Determine the exact information lattice and minimax envelopes when both carrier resolutions are oblique and only projector-condition and operator-norm budgets are declared.
2. **Incidence-locus stability.** Relate principal angles, small singular values, and carrier perturbations to quantitative distance from $\{(A, B) : AB = 0\}$ without identifying near incidence with exact route failure.
3. **Productwise propagation.** Establish conditions under which one-block bounds compose along a finite routed product or Paper XX survivor recursion without exponential loss.
4. **Semantic lifts.** Supplementary Technical Note S1 establishes one proportional-row contract for hitting time and stationary mass. Classify broader preservation contracts under which block intervals imply probability, mixing, or channel statements for finite Markov partitions and open quantum channels.

Appendix A: Computational Artifacts

All listed artifacts are available in the [RIME repository](#) under `experiments/paper25/`; short paths below are relative to that directory.

Surface	Role	Short path
diagnostic sources	transport, localized perturbation, and two-state Markov controls	<code>*.py</code>
retained evidence	exact certificates, bounded observations, and receipts	<code>results/</code>
validators	exact replay, producer replay, claim alignment, and release closure	<code>validation/</code>
evidence metadata	claim/evidence alignment and exact release inventory	<code>claim-surface-map.json;</code> <code>release-manifest.json</code>
Supplement S1	proportional-row Markov theorems, hostile matrices, and bounded audit	<code>notes/proportional_markov_</code> <code>semantic_lift/</code>
partial Lean surface	typed transport and scalar-margin declarations	<code>lean/</code>

The exact and bounded-observation receipts have separate scopes and neither occurs in its own transitive closure. They record local closure verification, not independent validation or proof of the manuscript theorems. Reproduction commands and exact artifact paths are maintained in the experiment README and release manifest.

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